ARJUN RAJ SINGH

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EDUCATION

UNIVERSITY OF ILLINOIS URBANA-CHAMPAIGN, URBANA-CHAMPAIGN, IL — *Expected Graduation May 2030*

- B.S. in Computer Engineering

LEIGH HIGH SCHOOL, SAN JOSE, CA — *Graduated June 2026*

- GPA: 4.5 weighted / 4.0 unweighted | ACT: 36 | National Merit Finalist
- 5 on all 13 AP exams taken: Calculus AB, Calculus BC, Chemistry, US History, Computer Science A, Physics 1, Physics C Mechanics, Statistics, Physics 2, Physics C E&M, Biology, US Government, English Literature

PROJECTS

PERSONAL CPU & COMPUTER SYSTEM — *July 2022 – Present*

- Self-learned Electrical Engineering and Computer Architecture
- Architected, designed, and simulated several computer processors on Logisim
- Hand-built the CPU on 40+ breadboards, module by module using 100+ IC chips
- Used precision testing instruments like an oscilloscope for troubleshooting
- Wrote automation scripts in Python to turn custom assembly code into machine code, and flash it onto the CPU
- Used my instruction set to write programs for the CPU, ranging from test programs demonstrating its functionality, Fibonacci Series, the game snake, and machine code monitor over VGA and keyboard
- Constructed fast, stable, and reliable interfaces—both hardware and software—to connect peripherals including a character display, LED matrix, keypad, custom keyboard, and VGA monitor

SYSTEMVERILOG FPGA IMPLEMENTATION OF MY CPU & COMPUTER SYSTEM — *June 2026 – Present*

- Reimplemented original custom breadboard CPU in synthesizable SystemVerilog using Vivado on a Xilinx Artix-7 FPGA (Basys 3)
- Preserved the original datapath, control logic, registers, ISA, opcodes, and execution behavior in a core with RAM and ROM
- Utilized onboard LEDs, buttons, switches, and displays for real-time bus, register, and data monitoring
- Achieved full hardware synthesis utilizing 9,577 LUTs and 171 FF
- Integrated LED-matrix and keypad peripherals through the FPGA I/O system
- Ran the original Snake binary unchanged on the FPGA implementation

DISCRETE-LOGIC SNAKE GAME — *Jan 2023 – Jun 2023*

- Designed snake game from discrete logic components in Logisim w/o microcontroller or programming
- Runs on 8x8 LED matrix, implementing complete game function directly in digital logic including movement, growth, collisions, and game state control
- Served as precursor to current CPU iteration through analyzing proposed workloads (snake) and useful ALU operations and memory addressing mechanisms

BIT-BANGED I2C EEPROM INTERFACE — *April 2026*

- Implemented I2C manually with Arduino digitalRead() and digitalWrite() for an AT24C256 serial EEPROM, including start/stop, addressing, acknowledgements, and byte read/write operations
- Verified contents with XGPro programmer

AVIATION

PRIVATE PILOT GLIDER (MEMBERSHIP CO-CHAIR) — *July 2021 – Present*

BAY AREA SOARING ASSOCIATES, HOLLISTER

- Logged 313 flights, 77 total flight hours, and 28 Pilot-In-Command (PIC) hours across high-performance sailplanes (ASK-21, SZD-51, DG505, DG1000)

PRIVATE PILOT AIRPLANE — *June 2024 – Present*

SPARROWHAWK AVIATION, HOLLISTER

- Logged 73.4 flight hours, 11 Pilot-In-Command (PIC) hours, and 175 landings across single-engine aircraft (Piper Archer, Cessna 152, Citabria)

RESEARCH & LEADERSHIP

COSMOS* SEMICONDUCTOR MATERIALS & DEVICE ENGINEERING — *July – August 2025*

UNIVERSITY OF CALIFORNIA, SANTA CRUZ

- Learned in-depth about the chemistry, physics, and manufacturing process of semiconductor devices such as transistors, LEDs, diodes, and solar cells
- Conducted group research and presented poster on EEPROM Architecture and FGMOS Physics

* California State Summer School for Mathematics & Science

DIGITAL ELECTRONICS CLUB — *August 2024 – May 2026*

LEIGH HIGH SCHOOL, SAN JOSE

- Founded club to teach digital logic and hardware design; authored schematics and curriculum guiding members through breadboard builds

SMALL-SCALE INTEGRATION STEPPING STONES VERILOG MEETUP — *June 2026 – Present*

- Authored article connecting small-scale integration to SystemVerilog workflows
- Designed and built demo circuits that are easy for beginners to assemble and understand

MOUNTAIN VIEW TECHNOLOGY SHOWCASE - *July 2026*

- Presented FPGA implementation of my CPU at Verilog Meetup booth, explaining journey from small-scale integration to modern engineering workflows
- Introduced the Verilog Meetup on the stage in front of a crowd, explaining the educational mission and my project

SKILLS

- SystemVerilog, Computer Architecture, Digital Circuit Design, FPGA Synthesis (Vivado), Oscilloscopes, Semiconductors, Python, Custom Assembly, C++, Java, Arduino, Logisim, 3D Printing, Ceramics